MCP Geo Documentation Set for the UK Public Sector AI Community

Repository Journey, Evaluation, and Delivery Readiness

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Chapter 1

Introduction

1.1 Purpose

This document package explains what was discovered and built in the `mcp-geo` repository for UK public-sector delivery audiences.

1.2 Audience

The text is designed for:

- novice readers new to MCP and geospatial AI integration,
- apprentices with early-stage coding knowledge,
- delivery and assurance reviewers.

1.3 Scope

The documented scope includes this repository's implementation, testing, troubleshooting, and release evidence. It excludes implementation internals of git submodules.

1.4 Reading Path

1. Origins and scope (Chapter 2)
2. System design (Chapter 3)
3. Timeline and external milestones (Chapter 4)
4. Harness and evaluation (Chapters 7 and 8)
5. BDUK extension and governance path (Chapter 9)

Chapter 2

Origin, Scope, and Acknowledgement

2.1 Initial Concept

The initial concept selected a broad read-only capability surface combining geospatial and statistical queries. This was treated as a practical stress test for MCP delivery in public-sector contexts [1, 2].

2.2 Acknowledgement of Early Contribution

The repository history includes an explicit linkage to early `os-mcp` work. The origin narrative also records acknowledgement of early contribution by Chris Carlon in initial project prompts captured by Codex reporting artifacts [3].

2.3 Change of Course

Delivery moved toward a fully tracked repository model with:

- durable context tracking,
- progress tracker workstreams,
- release-note discipline,
- test and troubleshooting evidence loops.

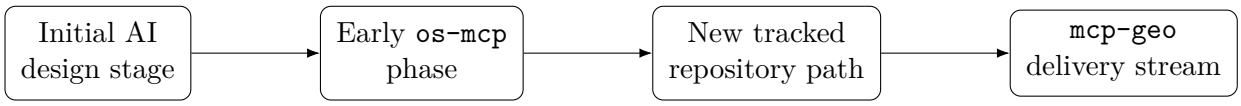

Figure 2.1: Observed project lineage represented in repository artifacts.

Chapter 3

System and Components

3.1 Runtime Structure

The system exposes MCP functionality through:

- HTTP transport (including `/mcp`),
- STDIO transport via JSON-RPC adapter,
- tool and resource catalogs with discovery and call endpoints.

3.2 Component Map

- `server/`: transports, routing, protocol handling
- `tools/`: domain handlers (OS, ONS, NOMIS, admin, map)
- `resources/`: static or generated resource payloads
- `tests/`: regression, compatibility, and harness validation

3.3 Current Functional Scale

Current registry evidence shows 81 tools loaded in the runtime registry.

3.4 Architecture Diagram

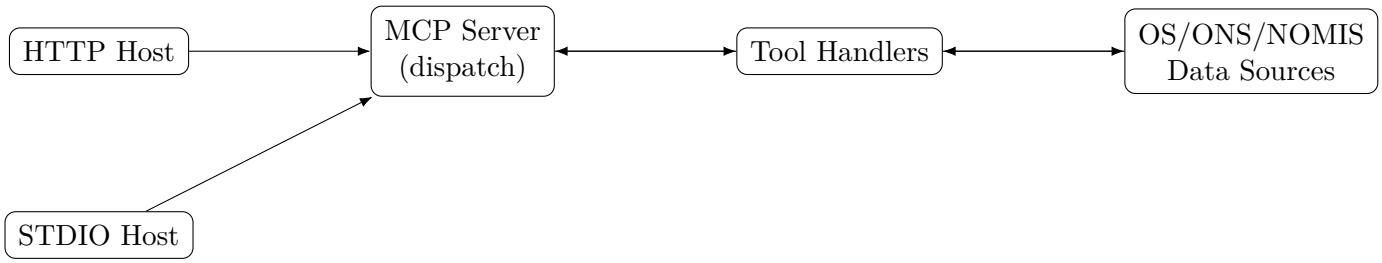

Figure 3.1: High-level runtime flow for user question execution.

Chapter 4

Timeline: Repository and Ecosystem

4.1 Internal Repository Milestones

Date	Milestone	Evidence
2025-08-20	Initial commit wave	Git history [4]
2025-09-17	Early release sequence and STDIO/ONS maturity	Release notes [5]
2026-01-20 to 2026-01-30	Rapid protocol, transport, and map/boundary delivery wave	Release notes [6, 7, 8]
2026-02-01 to 2026-02-25	Reliability, catalog, interop, and reporting expansions	Release notes and trackers [9, 10, 11]
2026-03-01 to 2026-03-03	Compact-window and troubleshooting hardening	Progress/changelog/troubleshooting [11, 12, 13]

4.2 External Milestones Affecting Delivery

Date	External event	Delivery impact
2025-03-26	MCP changelog introduces Streamable HTTP and auth framework updates [14]	Transport and protocol negotiation design
2025-04-03	VS Code 1.99 publishes MCP support in stable channel [15]	Host compatibility priority increased
2025-05-16	OpenAI introduces Codex [16]	Tooling surface evolution
2025-06-18	Anthropic announces remote MCP support in Claude Code [17]	Interop and diagnostics expanded
2025-11-25	MCP revision changelog publishes major auth/task updates [18]	Multi-version support and tracking hardened
2026-01-26	MCP Apps extension declared stable [19]	UI extension alignment work
2026-02-02	OpenAI introduces Codex app for macOS [20]	Desktop workflow integration emphasis

Chapter 5

Reproducible Development Environment

5.1 Baseline Toolchain

- Python 3.11+
- Node.js 20
- Docker for containerized workflows
- optional PostGIS service for boundary and cache pipelines

5.2 Devcontainer Path

The recommended reproducible setup uses `.devcontainer/devcontainer.json` and docker-compose services with post-create dependency provisioning.

5.3 Secrets and Runtime Configuration

The repository supports both direct environment keys and file-based key loading (`OS_API_KEY_FILE`) to reduce hardcoded secret handling in host configuration paths.

5.4 Core Verification Commands

- `pip install -e .[test]`
- `uvicorn server.main:app -reload`
- `pytest -q`
- `npm -prefix playground run test`

Further details are documented in repository runbooks [21, 22, 23, 24].

Chapter 6

Standards, Clients, and Tooling Evolution

6.1 Protocol Compatibility Posture

The repository documents a compatibility-first approach:

- preferred MCP protocol revision: 2025-11-25,
- compatibility support for 2025-06-18, 2025-03-26, and 2024-11-05,
- MCP Apps extension tracking for 2026-01-26.

This is evidenced in the protocol tracker and release notes [25, 26].

6.2 Client Interoperability Evolution

Observed host/client differences materially affected implementation. The project added targeted mitigations for:

- tool-name compatibility,
- startup payload size constraints,
- UI mount and runtime variance,
- transport framing and metadata expectations.

6.3 Tooling Surface Transition

Repository evidence shows a delivery transition from early Claude-focused interoperability work toward broader Codex-driven implementation throughput and reporting workflows [11, 27, 3].

Chapter 7

Harness, Permissions, and Diagnostics Journey

7.1 Execution Progression

The execution model progressed through three practical stages:

1. repo-local execution,
2. devcontainer execution,
3. host-integrated execution for real runtime diagnostics.

7.2 Diagnostics Loop

Figure 7.1: Repeatable troubleshooting loop used across incident classes.

7.3 Incident Families Addressed

- constrained-host tool discovery and naming mismatches,
- MCP-Apps runtime and bridge behaviors,
- map payload and startup pressure,
- NOMIS/ONS query-shape drift,
- collection aliasing and count semantics in feature queries.

The resulting evidence base is represented in troubleshooting reports and tracker workstreams [13, 28, 29].

Chapter 8

Effectiveness Evaluation

8.1 Method

Evaluation draws on repository-native metrics:

- release and delivery evidence,
- harness and regression outcomes,
- troubleshooting closure artifacts,
- residual blockers in active tracker records.

8.2 Measured Signals

- Tool registry scale: 81 tools (runtime registry load).
- Live harness closure recorded at 6900/6900 (100%) in tracker evidence.
- Repeated full-suite and compact-host validation cycles recorded in progress and changelog streams.

8.3 Residual Constraints

- external host runtime behavior remains a documented blocker for specific MCP-Apps pathways,
- CI automation remains a tracked gap,
- future controlled-data expansion requires formal RBAC/ABAC controls.

8.4 Codex Activity Metrics

Latest long-horizon snapshot (2026-03-04):

- 59 sessions,
- 668.3 active hours,
- 1.464B tokens,
- 16,588 tool calls.

Source: [3].

Chapter 9

BDUK Pilot Extension and Access-Control Requirements

9.1 Pilot-Readiness Workstreams

A practical pilot extension path requires:

1. governance and assurance gate,
2. access-control uplift,
3. operational hardening,
4. onboarding and support pathways.

9.2 RBAC and ABAC Requirement

As dataset sensitivity expands, policy controls need to move from prompt-level intent to enforced runtime checks.

- RBAC: permission by role,
- ABAC: permission by role + context + data attributes,
- deterministic allow/deny records with audit linkage.

9.3 Policy Integration Pattern

- identity propagation from host,
- policy decision point before tool execution,
- policy enforcement point in server logic,
- logged evidence for assurance review.

Chapter 10

Future Direction for MCP in UK Public Sector Use

10.1 Current Direction

The evidence indicates continued movement toward:

- stronger authorization and consent structures,
- broader host/client support with uneven runtime parity,
- stronger interoperability and auditability expectations.

10.2 Practical Implication

For UK public-sector delivery, MCP is most effective as part of a governed middle layer that combines:

- protocol compatibility,
- semantic consistency,
- policy enforcement,
- observability and incident discipline.

10.3 Near-Term Priorities

- complete CI gate automation,
- strengthen auth extension alignment,
- maintain compatibility across evolving hosts and protocol revisions,
- preserve trace-to-fix workflow discipline.

Chapter 11

Evidence Index

11.1 Core Repository Evidence

- `CONTEXT.md`
- `PROGRESS.MD`
- `CHANGELOG.md`
- `RELEASE_NOTES/`

11.2 Research and Reports

- Apps-to-Answers research pack [1]
- Deep research report [2]
- Map delivery and governance reports [30, 31]

11.3 Troubleshooting and Forensics

- master troubleshooting runbook [13]
- forensic report [32]
- NOMIS and peat failure analyses [29, 28]

11.4 External References

- MCP specification revisions and changelogs [14, 18, 33]
- MCP Apps stabilization announcement [19]
- host/client milestones for Claude, VS Code, and Codex [34, 17, 15, 16, 20]

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